2026A - Programación de sistemas avanzados

Ingeniería en Computación

Presentación

RUST • IN • PEACE

AVANCE

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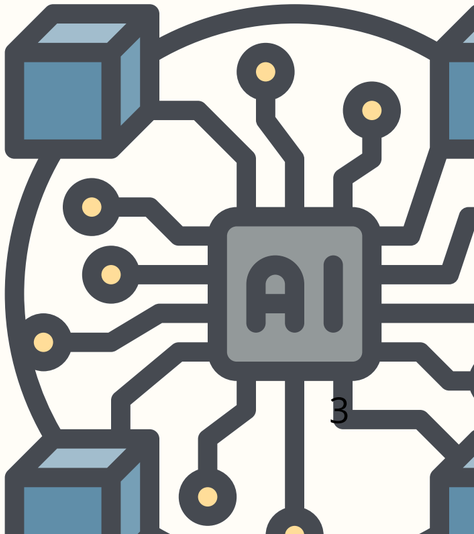
Introducción

En esta primera fase del proyecto se diseñó e implementó la infraestructura base necesaria para la ejecución de un sistema distribuido.

El trabajo consistió en la configuración manual de una red privada virtual (VPN) utilizando WireGuard bajo una arquitectura tipo hub-and-spoke, así como en la preparación de un entorno de contenedores Docker en cada nodo. Además de fundamentos técnicos para el desarrollo posterior de un algoritmo distribuido en Rust.

Objetivo del Proyecto

1. Construir VPN privada tipo hub-and-spoke.
2. Configurar contenedores Docker en cada nodo.
3. Preparar entorno para algoritmo distribuido en Rust.
4. Resolver problemas reales de red (NAT, firewall, ruteo).



Metodología usada

Se adoptó un modelo de Scrum ligero, organizando el proyecto en tres sprints enfocados en:

1 Sprint 1

Configuración y validación de la VPN.

2 Sprint 2

Implementación de la infraestructura Docker.

3 Sprint 3

Definición del algoritmo distribuido en Rust.

Feb 4, 2026 - Feb 4, 2026

Sprint 0

Definición de cantidad total de sprints del proyecto. Asignación y repartición de las actividades del primer sprint.

Feb 4, 2026 - Feb 9, 2026

Sprint 1

Repositorio creado con estructura base, generación de llaves de WireGuard, creación y levantamiento de VPN. Pruebas de conexión y resolución de problemas de VPN.

Feb 13, 2026 - Feb 16, 2026

Sprint 2

Creación y levantamiento de Docker.

Feb 16, 2026 - Feb 19, 2026

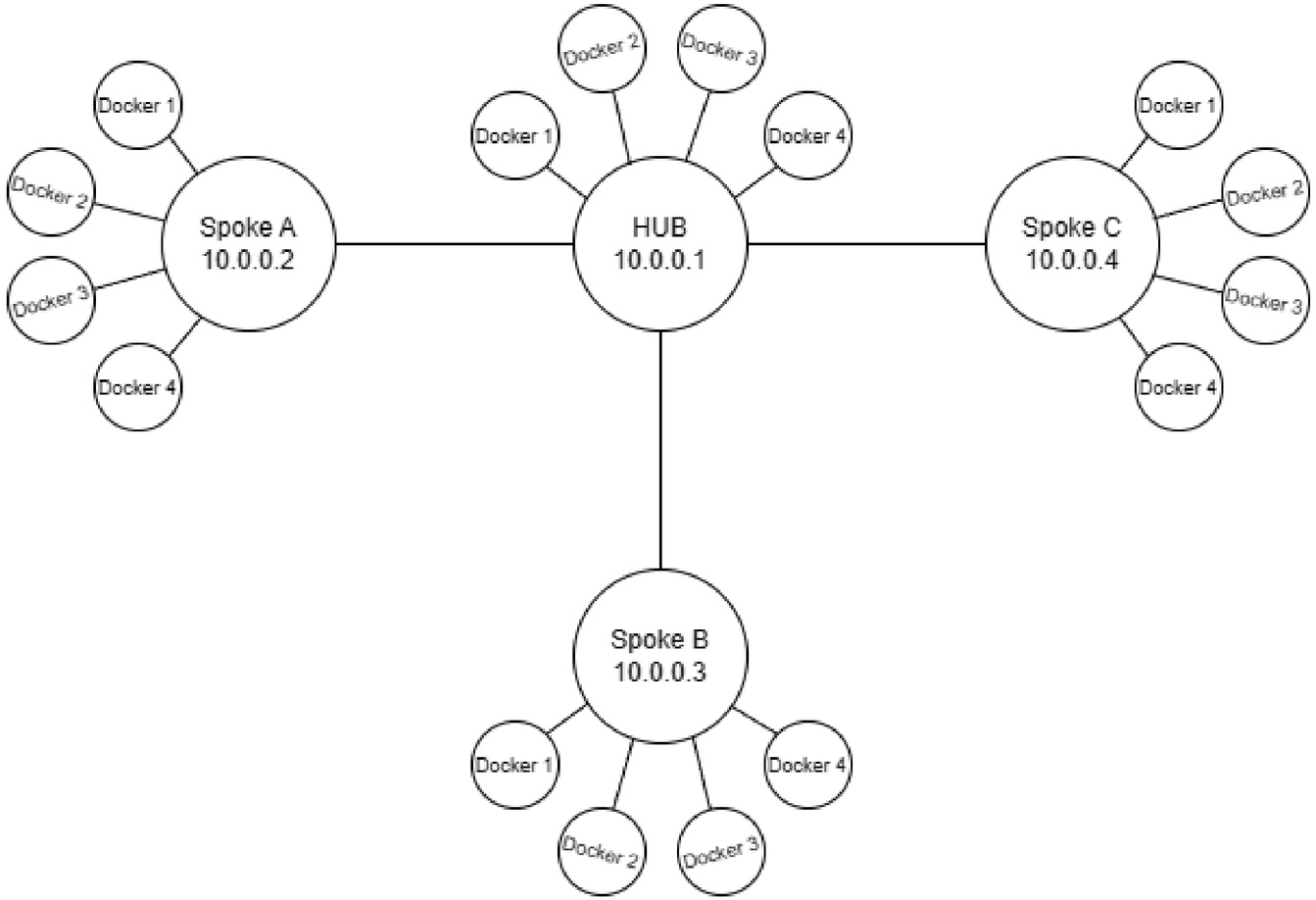
Sprint 3

Definición y justificación del problema computacional distribuido a implementar, estructura de proyecto, definición de mensajes, "hello distributed" verificable.

Reuniones en total:7

➔ Topología de la VPN.

- WireGuard (UDP 51820)
- Red privada: 10.0.0.0/24
- Modelo hub-and-spoke
- 1 Hub central
- 3 Spokes
- 4 contenedores por nodo
- Comunicación exclusivamente por VPN



Evidencia de ejecución, conexion y resultados.

Se validó el funcionamiento de la infraestructura mediante pruebas reales de conectividad y ejecución:

- **Estado activo del túnel VPN (wg show)**
- **Comunicación entre nodos mediante ping**
- **Contenedores activos en cada nodo (docker ps)**
- **Pruebas de comunicación entre contenedores vía red privada.**

Los resultados confirman que la infraestructura está operativa y lista para la ejecución del sistema distribuido.

Handshake

alvarogc@hub-server: ~

```
[Nodo-Alvaro] ~ $ sudo wg-quick up wg0
[#] ip link add wg0 type wireguard
[#] wg setconf wg0 /dev/fd/63
[#] ip -4 address add 10.0.0.1/24 dev wg0
[#] ip link set mtu 1420 up dev wg0
[#] sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
[Nodo-Alvaro] ~ $ sudo wg show
interface: wg0
  public key: 5zkHZoWZT1Wo5WZI5NwA18xwl9dfR2pb6QXWxPxDVSA=
  private key: (hidden)
  listening port: 51820

peer: AbfVXAPwyGJ87zyW4cQISezzMMJUCSw6JZ4vSXzkKXY=
  endpoint: 192.168.100.188:43493
  allowed ips: 10.0.0.3/32
  latest handshake: 37 seconds ago
  transfer: 212 B received, 92 B sent

peer: m61y9qWq/2t2iJ34AJlitbY1BJOn1SY2o1g/nfkWY1o=
  endpoint: 192.168.100.96:50498
  allowed ips: 10.0.0.2/32
  latest handshake: 46 seconds ago
  transfer: 212 B received, 92 B sent

peer: NWgnYISSv4oWytP5T3/xjPm3NT47hs6PJm2915u7ME0=
  endpoint: 192.168.100.196:46924
  allowed ips: 10.0.0.4/32
  latest handshake: 51 seconds ago
  transfer: 212 B received, 92 B sent
[Nodo-Alvaro] ~ $
```

mario-ubuntu@ubuntu-serve

```
[Nodo-Nora] ~ $ sudo wg-quick up wg0
[#] ip link add wg0 type wireguard
[#] wg setconf wg0 /dev/fd/63
[#] ip -4 address add 10.0.0.4/24 dev wg0
[#] ip link set mtu 1420 up dev wg0
[Nodo-Nora] ~ $ sudo wg show
interface: wg0
  public key: NWgnYISSv4oWytP5T3/xjPm3NT47hs6PJm2915u7ME0=
  private key: (hidden)
  listening port: 46924

peer: 5zkHZoWZT1Wo5WZI5NwA18xwl9dfR2pb6QXWxPxDVSA=
  endpoint: 192.168.100.186:51820
  allowed ips: 10.0.0.0/24
  latest handshake: 23 seconds ago
  transfer: 92 B received, 180 B sent
  persistent keepalive: every 25 seconds
[Nodo-Nora] ~ $
```

ivang@LAPTOP-KCVLMDHQ:

```
[Nodo-Andrea] ~ $ sudo wg-quick up wg0
[#] ip link add wg0 type wireguard
[#] wg setconf wg0 /dev/fd/63
[#] ip -4 address add 10.0.0.2/24 dev wg0
[#] ip link set mtu 1392 up dev wg0
[#] resolvconf -a wg0 -m 0 -x
[Nodo-Andrea] ~ $ sudo wg show
interface: wg0
  public key: m61y9qWq/2t2iJ34AJlitbY1BJOn1SY2o1g/nfkWY1o=
  private key: (hidden)
  listening port: 50498

peer: 5zkHZoWZT1Wo5WZI5NwA18xwl9dfR2pb6QXWxPxDVSA=
  endpoint: 192.168.100.186:51820
  allowed ips: 10.0.0.0/24
  latest handshake: 29 seconds ago
  transfer: 92 B received, 212 B sent
  persistent keepalive: every 25 seconds
[Nodo-Andrea] ~ $
```

guia — guia@ubuntu-server: ~ — ssh guia@192.168.100.188 — 120×30

```
[Nodo-Palma] ~ $ sudo wg-quick up wg0
[sudo] password for guia:
[#] ip link add wg0 type wireguard
[#] wg setconf wg0 /dev/fd/63
[#] ip -4 address add 10.0.0.3/24 dev wg0
[#] ip link set mtu 1420 up dev wg0
[#] resolvconf -a wg0 -m 0 -x
[Nodo-Palma] ~ $ sudo wg show
interface: wg0
  public key: AbfVXAPwyGJ87zyW4cQISezzMMJUCSw6JZ4vSXzkKXY=
  private key: (hidden)
  listening port: 43493

peer: 5zkHZoWZT1Wo5WZI5NwA18xwl9dfR2pb6QXWxPxDVSA=
  endpoint: 192.168.100.186:51820
  allowed ips: 10.0.0.0/24
  latest handshake: 26 seconds ago
  transfer: 92 B received, 212 B sent
  persistent keepalive: every 25 seconds
[Nodo-Palma] ~ $
```

Pings

alvarogc@hub-server: ~

```
[Nodo-Alvaro] ~ $
[Nodo-Alvaro] ~ $
[Nodo-Alvaro] ~ $ ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=5.46 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=6.60 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=5.15 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=5.45 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 5.152/5.666/6.604/0.555 ms
[Nodo-Alvaro] ~ $ ping -c 4 10.0.0.3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=78.9 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=104 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=64 time=7.45 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=64 time=45.5 ms

--- 10.0.0.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 7.450/58.884/103.709/36.168 ms
[Nodo-Alvaro] ~ $ ping -c 4 10.0.0.4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=6.53 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=6.48 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=6.87 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=64 time=6.02 ms

--- 10.0.0.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 6.016/6.473/6.865/0.302 ms
[Nodo-Alvaro] ~ $
```

mario-ubuntu@ubuntu-serve

```
[Nodo-Nora] ~ $ ping -c 4 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=8.79 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=6.76 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=7.31 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=5.87 ms

--- 10.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 5.870/7.182/8.785/1.058 ms
[Nodo-Nora] ~ $ ping -c 4 10.0.0.3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=63 time=64.7 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=63 time=86.8 ms
64 bytes from 10.0.0.3: icmp_seq=3 ttl=63 time=111 ms
64 bytes from 10.0.0.3: icmp_seq=4 ttl=63 time=30.7 ms

--- 10.0.0.3 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 30.669/73.299/110.977/29.549 ms
[Nodo-Nora] ~ $ ping -c 4 10.0.0.4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=64 time=0.033 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=64 time=0.056 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=64 time=0.031 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=64 time=0.044 ms

--- 10.0.0.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3064ms
rtt min/avg/max/mdev = 0.031/0.041/0.056/0.010 ms
[Nodo-Nora] ~ $
```

ivang@LAPTOP-KCVLMDHQ:

```
[Nodo-Andrea] ~ $
[Nodo-Andrea] ~ $ ping -c 4 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=5.00 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=5.60 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=5.84 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=5.36 ms

--- 10.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3001ms
rtt min/avg/max/mdev = 4.995/5.447/5.838/0.311 ms
[Nodo-Andrea] ~ $ ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=0.142 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=0.048 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=64 time=0.040 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=64 time=0.057 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3051ms
rtt min/avg/max/mdev = 0.040/0.071/0.142/0.041 ms
[Nodo-Andrea] ~ $ ping -c 4 10.0.0.4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=63 time=12.5 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=63 time=11.0 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=63 time=11.0 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=63 time=10.9 ms

--- 10.0.0.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3023ms
rtt min/avg/max/mdev = 10.876/11.342/12.495/0.667 ms
[Nodo-Andrea] ~ $
```

guia — guia@ubuntu-server: ~ — ssh guia@192.168.100.188 — 120×30

```
[Nodo-Palma] ~ $ ping -c 4 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=12.3 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=12.6 ms
64 bytes from 10.0.0.1: icmp_seq=3 ttl=64 time=7.20 ms
64 bytes from 10.0.0.1: icmp_seq=4 ttl=64 time=10.4 ms

--- 10.0.0.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3012ms
rtt min/avg/max/mdev = 7.202/10.619/12.623/2.154 ms
[Nodo-Palma] ~ $ ping -c 4 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=63 time=18.3 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=63 time=16.6 ms
64 bytes from 10.0.0.2: icmp_seq=3 ttl=63 time=17.3 ms
64 bytes from 10.0.0.2: icmp_seq=4 ttl=63 time=22.3 ms

--- 10.0.0.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3016ms
rtt min/avg/max/mdev = 16.635/18.619/22.264/2.183 ms
[Nodo-Palma] ~ $ ping -c 4 10.0.0.4
PING 10.0.0.4 (10.0.0.4) 56(84) bytes of data.
64 bytes from 10.0.0.4: icmp_seq=1 ttl=63 time=15.4 ms
64 bytes from 10.0.0.4: icmp_seq=2 ttl=63 time=32.5 ms
64 bytes from 10.0.0.4: icmp_seq=3 ttl=63 time=13.5 ms
64 bytes from 10.0.0.4: icmp_seq=4 ttl=63 time=14.6 ms

--- 10.0.0.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3006ms
rtt min/avg/max/mdev = 13.478/19.027/32.549/7.837 ms
```

Infraestructura de contenedores Docker

Cada nodo ejecuta 4 contenedores Docker que actuarán como unidades de cómputo distribuido (workers).

La infraestructura se estandarizó mediante docker-compose.yml, garantizando reproducibilidad y consistencia entre nodos.

Se configuró network_mode: host para permitir comunicación directa sobre la red privada WireGuard, evitando sobrecarga de redes bridge internas.

Compose File

```yaml
services:
  worker1:
    build:
      context: ../
      dockerfile: docker/Dockerfile
    container_name: worker1
    network_mode: host
    restart: unless-stopped

  worker2:
    build:
      context: ../
      dockerfile: docker/Dockerfile
    container_name: worker2
    network_mode: host
    restart: unless-stopped

  worker3:
    build:
      context: ../
      dockerfile: docker/Dockerfile
    container_name: worker3
    network_mode: host
    restart: unless-stopped

  worker4:
    build:
      context: ../
      dockerfile: docker/Dockerfile
    container_name: worker4
    network_mode: host
    restart: unless-stopped

```

```dockerfile
FROM rust:latest AS builder
WORKDIR /app
COPY rust/ ./rust/
WORKDIR /app/rust
RUN cargo build --release --bin worker

FROM ubuntu:24.04
RUN apt update && apt install -y libssl-dev ca-certificates curl iputils-ping && rm -rf /var/lib/apt/lists/*
COPY --from=builder /app/rust/target/release/worker /usr/local/bin/worker

CMD ["worker"]
```

Docker ps: evidencia de cuatro contenedores

```
ivang@LAPTOP-KCVLMDHQ:
[Nodo-Andrea] ~/proyecto-rin/docker $ docker ps
CONTAINER ID   IMAGE            COMMAND    CREATED              STATUS          PORTS     NAMES
ec3f29cb59d1   docker-worker3   "worker"   About a minute ago   Up 24 seconds             worker3
802083a84468   docker-worker1   "worker"   About a minute ago   Up 24 seconds             worker1
f9909caf0760   docker-worker4   "worker"   About a minute ago   Up 24 seconds             worker4
73755c492f23   docker-worker2   "worker"   About a minute ago   Up 24 seconds             worker2
[Nodo-Andrea] ~/proyecto-rin/docker $
```

Pruebas de comunicación entre contenedores

```
root@hub-server: /        Windows PowerShell

 ✔Container worker2     Created                                                    0.5s
 ✔Container worker3     Created                                                    0.4s
alvarogc@hub-server:~/proyecto-rin/workers$ docker ps
CONTAINER ID   IMAGE             COMMAND            CREATED         STATUS         PORTS     NAMES
fce1acfa5414   workers-worker4   "sleep infinity"   4 minutes ago   Up 4 minutes             worker4
029c3423db0f   workers-worker3   "sleep infinity"   4 minutes ago   Up 4 minutes             worker3
8cc3df9a614f   workers-worker1   "sleep infinity"   4 minutes ago   Up 4 minutes             worker1
ca26127fa470   workers-worker2   "sleep infinity"   4 minutes ago   Up 4 minutes             worker2
alvarogc@hub-server:~/proyecto-rin/workers$ docker exec -it worker1 bash
root@hub-server:/# ping -c 2 10.0.0.1
PING 10.0.0.1 (10.0.0.1) 56(84) bytes of data.
64 bytes from 10.0.0.1: icmp_seq=1 ttl=64 time=0.046 ms
64 bytes from 10.0.0.1: icmp_seq=2 ttl=64 time=0.128 ms

--- 10.0.0.1 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1019ms
rtt min/avg/max/mdev = 0.046/0.087/0.128/0.041 ms
root@hub-server:/# ping -c 2 10.0.0.2
PING 10.0.0.2 (10.0.0.2) 56(84) bytes of data.
64 bytes from 10.0.0.2: icmp_seq=1 ttl=64 time=4.44 ms
64 bytes from 10.0.0.2: icmp_seq=2 ttl=64 time=4.24 ms

--- 10.0.0.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 4.238/4.337/4.437/0.099 ms
root@hub-server:/# ping -c 2 10.0.0.3
PING 10.0.0.3 (10.0.0.3) 56(84) bytes of data.
64 bytes from 10.0.0.3: icmp_seq=1 ttl=64 time=74.4 ms
64 bytes from 10.0.0.3: icmp_seq=2 ttl=64 time=96.7 ms

--- 10.0.0.3 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1002ms
rtt min/avg/max/mdev = 74.378/85.546/96.714/11.168 ms
root@hub-server:/#
```

Hub worker
1 > peers

Pruebas de comunicación entre contenedores

```
alvarogc@hub-server: ~/proy
alvarogc@hub-server:~/proyecto-rin/workers$ docker exec -it worker4 bash
root@hub-server:/# nc 10.0.0.2 5000
```

```
root@LAPTOP-KCVLMDHQ: /
ivang@LAPTOP-KCVLMDHQ:~/proyecto-rin/workers$ docker exec -it worker2 bash
root@LAPTOP-KCVLMDHQ:/# nc -lvnp 5000
Listening on 0.0.0.0 5000
Connection received on 10.0.0.1 49350
```

Nodo HUB
worker 4 >
Nodo Peer2
worker2

PLANTEAMIENTO DEL ALGORITMO DISTRIBUIDO EN RUST

Se implementará el cálculo distribuido del Conjunto de Mandelbrot bajo un modelo Coordinador–Workers.

Coordinador (Hub):

Mantiene una cola de tareas y asigna trabajo bajo demanda

Workers (Peers):

Solicitan una tarea → procesan el bloque → envían resultado → solicitan la siguiente.

• tokio •axum •reqwest • serde y serde_json

alvarogc@hub-server: ~

```
[Nodo-Alvaro] ~/proyecto-rin/rust $ cargo run --bin coordinator
    Finished `dev` profile [unoptimized + debuginfo] target(s) in 1.45s
     Running `target/debug/coordinator`
Coordinador iniciado. Escuchando peticiones en http://0.0.0.0:3000
```

ivang@LAPTOP-KCVLMDHQ:

```
[Nodo-Andrea] ~/proyecto-rin/docker $ docker compose up -d --build
[+] Building 1.6s (14/18)
 => => transferring context: 2B                                                                    0.0s
 => => reading from stdin 1.77kB                                                                   0.0s
 => [worker1 internal] load build definition from Dockerfile                                       0.0s
 => => transferring dockerfile: 374B                                                               0.0s
 => [worker2 internal] load metadata for docker.io/library/ubuntu:24.04                            1.1s
 => [worker3 internal] load metadata for docker.io/library/rust:latest                             1.0s
 => [worker1 internal] load .dockerignore                                                          0.0s
 => => transferring context: 2B                                                                    0.0s
 => [worker1 stage-1 1/3] FROM docker.io/library/ubuntu:24.04@sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c  0.1s
 => => resolve docker.io/library/ubuntu:24.04@sha256:d1e2e92c075e5ca139d51a140fff46f84315c0fdce203eab2807c7e495eff4f9  0.1s
 => [worker2 internal] load build context                                                          0.0s
 => [worker4 builder 1/5] FROM docker.io/library/rust:latest@sha256:80302520b7199f0504975bca59a914015e9fee088f759875dbbc23  0.1s
 => => resolve docker.io/library/rust:latest@sha256:80302520b7199f0504975bca59a914015e9fee088f759875dbbc238ca9509ee1  0.1s
```

PLANTEAMIENTO DEL ALGORITMO DISTRIBUIDO EN RUST

Qué falta implementar

- Integrar la lógica matemática de Mandelbrot en el worker.
- Permitir que procese coordenadas enviadas por el coordinador.
- Conexión de más de una maquina.

Qué se probará primero

- Envío y recepción correcta de un fragmento (una fila de píxeles) vía WireGuard.
- Validar que el tamaño de los datos no afecte la transferencia HTTP.

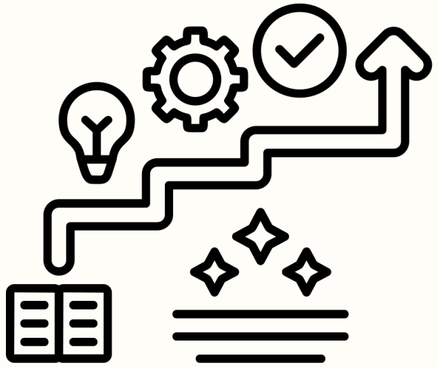
Cómo se medirá el rendimiento

- Medición del tiempo total usando std::time::Instant.
- Comparación con 1, 2, 3 y 4 nodos activos.

Prueba de tolerancia a fallos

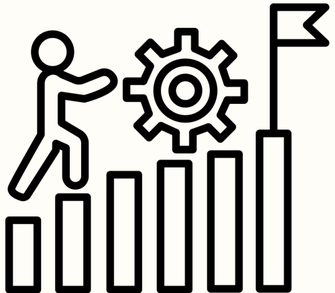
- Detener un worker durante la ejecución.
- Verificar que el coordinador detecte el timeout y reasigne la tarea.

Principales retos técnicos y aprendizajes.



Retos técnicos

- Redes detrás de CGNAT (sin IP pública).
- Limitaciones de WSL2 para tráfico UDP.
- Configuración de firewall y apertura de puertos.
- Diagnóstico de handshake en WireGuard.
- Configuración de ip_forward y ruteo interno.



Aprendizajes

- Implementación manual de una VPN funcional.
- Diagnóstico de problemas de red en entornos reales.
- Importancia del control del kernel y modo bridge.
- Integración entre red privada y contenedores Docker.
- Preparación de infraestructura para sistemas distribuidos.

Muchas

GRACIAS!